Highlights

Accuracy of open-data versus operator real-time bus arrival predictions: a paired evaluation of the Bide+ method in Biscay, Spain

Yeray Garrido

- Operator real-time bus predictions compared with an open-data method, same buses
- 140,158 paired predictions at 59 stops over five days in Biscay, Spain
- Within ±2 min: timetable 16%, operator real time 39%, open-data method 66%
- The open-data method is ahead from three minutes before arrival onwards
- Its learned segment travel times explain the gap; without them it falls behind

Accuracy of open-data versus operator real-time bus arrival predictions: a paired evaluation of the Bide+ method in Biscay, Spain

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Abstract

This paper compares an operator's real-time bus predictions with an independent predictor on the same buses and moments, for Bizkaibus, the interurban bus network of Biscay (Spain), which publishes vehicle positions and its timetable as open data and shows its own waiting times in an official app. For one week in November 2026, the official waiting times of 60 stops were recorded every minute with the position feed every 20 seconds. Each official announcement was matched to the bus it referred to, confirmed by the distance the service reports, and compared with the published timetable and with the Bide+ method, which predicts arrivals from the open data alone and was frozen before the week; all three were checked against the actual passage reconstructed from the feed. Over 140,158 paired predictions on five normal days, the timetable was within two minutes of the actual passage in 16% of cases, the official real time in 39% and Bide+ in 66%. The official real time was slightly more accurate below three minutes; from three minutes on, Bide+ was more accurate, by 32 to 36 percentage points at 10–20 minutes (95% intervals by trip), at 58 of 59 stops and at every hour; it had more very large errors at short horizons. An ablation attributes its advantage to learned segment travel times. The results apply to buses whose passage could be observed and confirmed (16% of announcements), use a ground truth from the same feed, and cover one week; data and code are open.

Keywords: bus arrival prediction, real-time passenger information, automatic vehicle location, paired forecast evaluation, open data, GTFS, SIRI

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1. Introduction

Real-time passenger information (RTPI) shortens the time passengers wait at stops and makes that wait less uncertain (Watkins et al. 2011; Dziekan and Kottenhoff 2007; Brakewood and Watkins 2019), but only when it is accurate: wrong predictions erode trust and can make passengers miss the bus they were told to expect (Gooze et al. 2013; Rahman et al. 2013). Most passengers receive these predictions from the operator, through its app and stop displays, or from journey planners that relay the operator's figures. Whether a third party with only the operator's *open* data can predict better than the operator is rarely measured, and we found no study that does so on the same buses and at the same moments.

This paper makes that measurement for Bizkaibus, the interurban bus network of the province of Biscay (Spain). Bizkaibus shows its own real-time waiting times in an official app, launched in 2023, and on some stop displays. It also publishes, as open data, the position of every vehicle in service, obtained by the Global Positioning System (GPS) and published as a SIRI Vehicle Monitoring feed (Service Interface for Real-time Information, CEN 2022), and its timetable in the General Transit Feed Specification (GTFS, MobilityData 2026), but not its predictions. The Bide+ method predicts arrivals from those two open sources alone. It is documented in a technical report (Garrido 2026a) and specified for reimplementation in a companion paper (Garrido 2026b), both of which evaluated it against the published timetable.

The research question is:

> *For the same bus, at the same stop and at the same minute, how close to the actual passage are the published timetable, the real-time waiting time announced by Bizkaibus and the arrival predicted by the Bide+ method?*

For one week in November 2026 the official waiting times of 60 stops were recorded every minute, together with the live feed every 20 seconds. Each official announcement was paired with the bus it referred to, confirmed by the distance that the official service gives, with that bus's published time at the stop, with the Bide+ prediction for that bus at that minute, and with the moment the bus actually passed.

The contributions are:

1. A paired protocol to compare a third-party predictor with an operator's real-time predictions when the operator publishes positions but not predictions: what to capture, how to match each announcement to a vehicle and confirm it, how to reconstruct the actual passage and screen out failed reconstructions, and how to report coverage and selection.
2. The comparison of the three sources over 140,158 paired predictions on five normal days, by actual and announced time to arrival, with inference that accounts for dependence within trips and within stops, a passenger decision model, an ablation of the learned component of Bide+, and 17 sensitivity analyses.
3. The capture program, the raw data and the analysis, openly available, so that the comparison can be repeated on other days or networks.

Section 2 reviews related work, Section 3 describes the three sources, Section 4 the data, Section 5 the comparison design, Section 6 the results, Section 7 their interpretation and limitations, and Section 8 concludes.

2. Related work

Arrival prediction from automatic vehicle location. Bus arrival prediction from automatic vehicle location (AVL) data has a long history. Early methods combined the last observed position with historical travel times between stops or timepoints (Lin and Zeng 1999; Cathey and Dailey 2003). Later work used Kalman filters (Cathey and Dailey 2003; Shalaby and Farhan 2004; Vanajakshi et al. 2009), artificial neural networks (Chien et al. 2002; Jeong and Rilett 2005) and, for stops served by several routes, models that share information across routes (Yu et al. 2011). Mazloumi et al. (2010) documented how travel time varies by time of day and segment, the variation that segment-based methods learn. Reviews (Moreira-Matias et al. 2015; Singh and Kumar 2022; Kumar et al. 2025) compare methods with each other or with the timetable. The method evaluated here belongs to the family of segment-based historical methods (Lin and Zeng 1999; Cathey and Dailey 2003): it learns the travel time of each segment between consecutive stops by time band, with shrinkage towards the all-day value, and adds the position of the vehicle.

Accuracy of operator real-time information. Evaluations of the predictions that operators actually show passengers are rare. Cats and Loutos (2016) evaluated the bus arrival information of Stockholm against AVL records and found that errors grow with the prediction horizon and depend on the

stop and the time of day. We found no study that compares an operator's live predictions with an independent predictor on the same vehicles and moments.

Value of accurate information for passengers. Real-time information reduces actual and perceived waiting times (Watkins et al. 2011; Dziekan and Kottenhoff 2007; Mishalani et al. 2006; Liu and Miller 2020), is valued by passengers (Ferris et al. 2010; Rahman et al. 2013) and can increase ridership (Tang and Thakuriah 2012; Brakewood et al. 2015); Brakewood and Watkins (2019) review these benefits. They depend on accuracy: wrong predictions reduce trust and lead to missed buses (Gooze et al. 2013).

Third-party predictors and open data. Predictors built only on public data are feasible: García-Mauriño et al. (2024) improve arrival predictions using only data available through public transport APIs, and Wessel et al. (2017) reconstruct actual passage times from a public vehicle location feed, the kind of ground truth used here.

Comparing predictors. Comparing two predictors on the same targets is a classic forecasting problem. The loss differential should be tested with errors that account for dependence (Diebold and Mariano 1995), the scoring rule should match the decision it serves (Gneiting 2011), and evaluation must be out of sample (Tashman 2000), with no information from the test period leaking into the predictor (Kaufman et al. 2012).

3. The three sources

3.1. The published timetable

Bizkaibus publishes its timetable as GTFS. Its intermediate times are interpolated at constant speed between the terminals, not observed, and on this network they drift from the actual passage as the trip advances: after the first hour of a trip most buses pass a stop more than two minutes before the published time (Garrido 2026b). The timetable is what a passenger sees in printed and online timetables and in any app without real-time data. Its prediction for a bus is the published time of its trip at the stop.

3.2. The official Bizkaibus real-time waiting times

The official app and some stop displays show, for each stop, the waiting time of the next buses of each line. The app obtains them from a public web service (`GetPasoParadaMobile_JSON`) that, for a stop, returns the next

two buses of every line that serves it, each with its waiting time in whole minutes, its distance to the stop in metres and a type field (empty in every response captured). How these times are computed is not published.

3.3. The Bide+ method

The Bide+ method (Garrido 2026a; Garrido 2026b) predicts the arrival of each bus at each stop from the public position feed and the timetable (Figure 1). Its main elements are:

- **Timetable time.** Each vehicle is matched to its scheduled trip, and its GPS position is projected onto the route and converted into the scheduled time at that point, τ. The arrival at a later stop with published time a_s is then built from the timetable difference $a_s - \tau$, which remains informative even when the absolute intermediate times are not.

- **Learned segment times.** The actual travel time of each segment between consecutive stops is learned from passages captured on previous days, by time band, as a median shrunk towards the all-day median of the segment in proportion to the number of observations; segments without data use a ratio by segment length. The learned time replaces the timetable difference of each segment ahead of the vehicle.

- **Three layers.** The prediction uses, in order, the projected GPS position, the last stop reported by the feed, and the timetable corrected trip by trip. Each layer abstains under explicit rules, and the state of the prediction (`live`, `estimated`, `departed`, ...) records which one was used.

- **Daily recalibration.** In production the tables are recalibrated every night, and a change is adopted only if it reduces the error on held-out data. In this paper they are frozen (Section 5).

The report (Garrido 2026a) gives every constant and the measurements behind it. The production engine is not public; an open reference implementation in Python agrees with it on all 407 test vectors produced by that engine, and it is the implementation used here. Against the published timetable, over 9.1 million predictions in 23 days, the method was within ±2 minutes in 92–96% of cases with the bus less than three minutes away and in 67% at 10 to 20 minutes, against 23–25% for the timetable (Garrido 2026b).

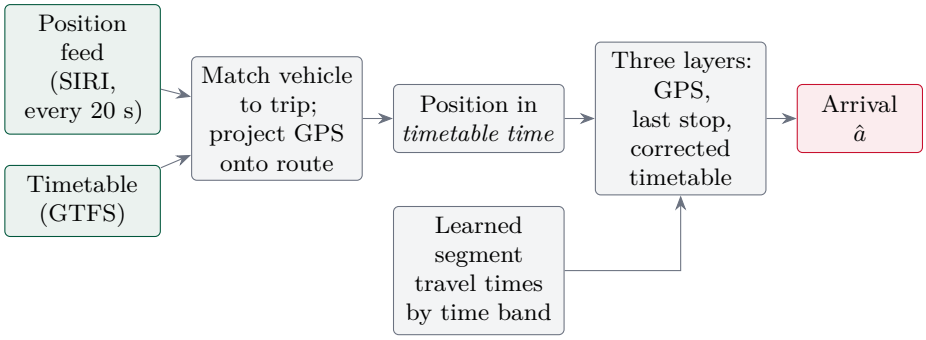


Figure 1: Outline of the Bide+ method (Garrido 2026a; Garrido 2026b). Only the operator's open data enter it: the position feed and the timetable.

4. Data

4.1. What was captured

A program running on GitHub Actions captured, from 06:00 to 23:00 Madrid time:

- **The live feed**, every 20 seconds: the public SIRI Vehicle Monitoring file of Bizkaibus, with the GPS position, the trip and the next stop of every vehicle in service. The file changes every 60 seconds.
- **The official waiting times**, about once a minute for each of 60 stops: the full response of the official web service, queried stop after stop (about one request per second in total), with the time t at which each request was sent and the response latency (median 0.9 s).
- **The timetable** of the day (GTFS), once a day.

4.2. Stops

The 60 stops, 28 of them *priority* stops, were chosen before the capture. The 32 other stops are intermediate stops with many buses and lines, at least 15 minutes after the start of the trip by median scheduled time, whose name in the official service matches the timetable. The 28 priority stops are stops suggested by passengers (in San Juan, Durango and Elorrio) and the Bilbao stops where, in earlier captures, the actual passage deviated most from the published time. Because this second criterion could favour a method that corrects the timetable, results are also given without the priority stops. Each stop and the reason for its choice are listed in the repository.

Table 1: Days of the capture (Madrid time). The feed and the official service were captured from about 06:00 to 23:00, except as noted.

Date	Day	Coverage
8 Nov 2026	Sunday	Complete
9 Nov 2026	Monday	12:42–23:00 (morning block lost: the capture overran its time limit)
10 Nov 2026	Tuesday	Complete
11 Nov 2026	Wednesday	Complete, *analysed separately*: general strike, minimum services (about 30%)
12 Nov 2026	Thursday	Feed complete; official times until 17:12 (the service stopped answering over HTTP)
13 Nov 2026	Friday	Complete
14 Nov 2026	Saturday	From 09:41; to be added in the next version

4.3. Days

The week runs from Sunday 8 to Saturday 14 November 2026 (Table 1). Wednesday 11 was a general strike with minimum services; it is analysed separately. Monday 9 lacks its morning (the capture overran the time limit of its job), and Thursday 12 lacks official times after 17:12 (the service stopped answering over HTTP; the capture has used HTTPS since). This version analyses the five complete or partial normal days from Sunday 8 to Friday 13; Saturday 14, captured from 09:41, will be added in the next version. The feed has no gap longer than two minutes on any day.

5. Comparison design

5.1. The three predictions

For a query sent at time t to stop s, and the bus matched to it (Section 5.3):

- **Timetable:** $\hat{a}_{\mathrm{tt}}$, the published time of the bus's trip at s.
- **Official:** $\hat{a}_{\mathrm{off}} = t + 60\,m$, where m is the announced waiting time in minutes. Whether the service rounds or truncates m is unknown; truncation is tested in the sensitivity analysis.
- **Bide+:** the prediction of the reference implementation in the last copy of the feed before t, at most 60 seconds old (median age 28 s), with the learned tables of Saturday 7 November, calibrated only with data from before the week and not updated during it. It is rounded to whole

minutes from t as the app shows it, $\hat{a}_{\mathrm{B+}} = t + 60\max(0, [(\hat{a} - t)/60])$, with halves rounded away from zero.

5.2. The actual passage

The actual passage a of a bus at a stop is reconstructed from the feed as in the companion paper: the midpoint between the last copy in which the next stop of the bus is s or an earlier one and the first copy in which it is a later one, without the bus changing trip. Since the feed changes every 60 seconds, a is known to within about ± 30 s. This definition uses the same feed field that Bide+ uses, so a systematic offset between it and the physical arrival would not affect the two sources equally; the sensitivity analysis therefore shifts a by ± 30 and ± 60 s.

The reconstruction can fail where the feed reports the next stop incorrectly. Pairs in which reaching the stop by a would require travelling at more than 30 m/s from the distance reported for the bus (by the feed or the official service) are discarded as failed reconstructions (0.6% of pairs, mostly at two stops near Bilbao terminals). One stop, Atxuri (3394), where all three sources are off by the same 17 to 30 minutes, is excluded entirely.

5.3. Matching announcements to buses

The official service does not identify the vehicle it announces. For a query at t to stop s, its first announcement for line ℓ is matched to the first bus of line ℓ, among those tracked in the feed, that actually passes s after t, and its second announcement to the second. The match is accepted only if the distance that the official service gives for the announced bus agrees with the distance along the route of the matched bus in the feed: within 500 m, or between 0.6 and 2 times it. This identity check removes announcements that refer to another bus, for example one that is not tracked or that the official service still shows after it has passed. Lines listed twice in the same response are left out (there were none), as are announcements whose matched bus passes more than 30 minutes after t.

When the matched bus is not in the feed at t, Bide+ can only give its fallback prediction (the corrected timetable). These pairs cannot pass the identity check, so they are excluded from the main analysis and included in a sensitivity analysis.

5.4. Metrics and inference

The error of each prediction is $\hat{a} - a$: positive when the bus arrives *earlier* than predicted. The main scoring rule is the share of predictions within ± 2 minutes, chosen in advance as in the companion paper because it matches

what a passenger can act on (Gneiting 2011). The median and mean absolute errors, the shares within ±1 and ±3 minutes, the median signed error and the share of buses that arrive more than two minutes early are also reported. Results are given by the actual time to arrival $a - t$ and by the announced time m.

Predictions of the same trip, and of the same stop, are not independent: one passage contributes a prediction every minute. Confidence intervals for the difference in the share within ±2 minutes are percentile intervals from 2,000 bootstrap resamples of whole trips (day, line, trip number and vehicle) and, separately, of whole stops (Efron and Tibshirani 1993). The mean loss differential $d = |e_{\text{off}}| - |e_{\text{B+}}|$ is tested with standard errors clustered by stop, in the manner of Diebold and Mariano (1995). With five days, day-level variance cannot be estimated.

Passenger model. To translate errors into consequences, a passenger is assumed to reach the stop k minutes before the predicted arrival ($k = 1, 2, 3$). The passenger misses the bus if it passes before that time; otherwise the wait is $a - (\hat{a} - k)$.

6. Results

6.1. From announcements to pairs

Table 2 follows the official announcements to the pairs analysed. Of the official announcements of the next bus less than 30 minutes ahead on the five normal days, 22.7% could be paired with a bus whose actual passage was observed, and 16.3% passed the identity check and entered the main analysis. Most announcements cannot be paired because the bus is near the end of its trip at the stop (the feed stops reporting it when the trip ends), is not tracked, or passes more than 30 minutes later. The main analysis has 140,158 pairs: 7,925 passages of 2,568 trips, made by 313 vehicles on 72 lines, at 59 stops.

6.2. Accuracy by time to arrival

The published timetable is within two minutes of the actual passage in 14–19% of cases at every horizon, with a median error of seven to nine minutes (Table 3, Figure 2). With the bus less than three minutes away, the official real time is the most accurate source: 92% of its announcements are within two minutes, against 90% for Bide+ (difference −2.4 to −0.8 points, 95% interval by trip). From three minutes on, Bide+ is more accurate: 79% against 55% at 3–6 minutes, 74% against 39% at 6–10, 62% against 28% at

Table 2: From official announcements to the pairs analysed, by day. *No passage*: announcements (first and second bus) whose bus has no observed passage within 30 minutes. *Paired*: announcements matched to a bus with an observed passage, $0 \leq a - t < 30$ min. *Fallback*: the bus was not in the feed at t (Bide+ gives its fallback prediction). *Excl.*: excluded stop or failed reconstruction. *Not conf.*: distance check failed.

Day	Queries	Announc.	Dupl.	No passage	Paired	Fallback	Excl.	Not conf.	Main
Sun 8 Nov	58,320	394,247	0	362,015	32,228	7,408	135	1,748	22,937
Mon 9 Nov	34,524	325,710	0	287,224	38,481	6,556	339	3,720	27,866
Tue 10 Nov	74,604	543,150	0	493,466	49,682	10,978	297	4,752	33,655
Wed 11 Nov (strike)	58,481	374,546	0	340,499	34,041	9,710	264	2,695	21,372
Thu 12 Nov	35,549	371,455	0	327,859	43,592	9,299	287	4,352	29,654
Fri 13 Nov	46,228	447,096	0	407,460	39,623	9,319	421	3,837	26,046

Table 3: Errors of the three sources by time to the actual passage. *Med.*: median absolute error (s). *±2*: share within two minutes (%). *Early*: share of buses arriving more than two minutes before the predicted time (%; error $\hat{a} - a > 120$ s).

		Timetable			Official real time			Bide+		
Minutes	Pairs	Med.	±2	Early	Med.	±2	Early	Med.	±2	Early
0–3	17,125	406	19	54	40	92	8	38	90	7
3–6	17,530	423	18	56	106	55	23	53	79	9
6–10	21,869	448	17	58	157	39	37	66	74	12
10–20	47,469	495	15	62	230	28	52	90	62	15
20–30	36,165	540	14	64	317	21	56	128	48	16
All	140,158	478	16	60	170	39	41	78	66	13

10–20 and 48% against 21% at 20–30 minutes. Over all horizons the shares are 16% for the timetable, 39% for the official real time and 66% for Bide+. Table 4 gives the shares within one minute and the median signed errors.

6.3. Inference

Table 5 gives the intervals of the difference in the share within two minutes, clustered by trip and by stop, and the paired test of the absolute error. Clustering by stop widens the intervals but does not change any conclusion: from three minutes on, the difference is positive with either clustering. The test of the absolute error tells a more nuanced story. Below six minutes the official real time has the smaller mean absolute error, because Bide+ has more large errors (Section 6.5); between 6 and 10 minutes the two do not differ; from 10 minutes on Bide+ has the smaller mean absolute error, by 93 to 160 s.

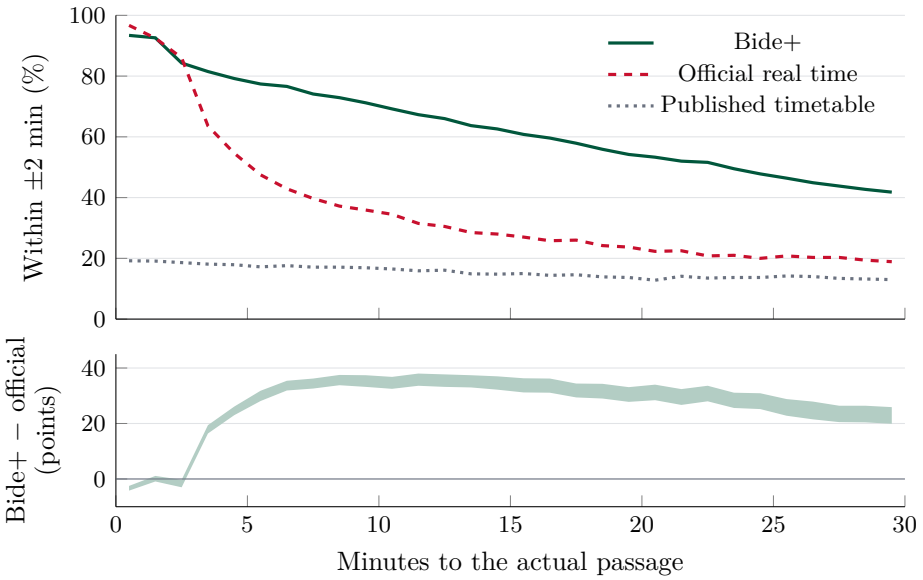


Figure 2: Top: share of predictions within two minutes of the actual passage, by minutes to that passage (one-minute bins). Bottom: 95% bootstrap interval (by trip) of the difference, Bide+ minus official, in percentage points. Main analysis, 140,158 pairs.

6.4. Accuracy by announced waiting time

Binning by the actual time to arrival conditions on the outcome. A passenger sees the announced time instead. Table 6 bins the same pairs by the minutes announced by the official service. The ranking is the same: when the official service announced 6 to 9 minutes, its announcement was within two minutes in 51% of cases and Bide+ in 76%; at 10 to 19 minutes, 35% and 66%. When it announced 30 minutes or more for a bus that came within 30 minutes (13% of the pairs, all with the identity confirmed), it was almost never within two minutes.

6.5. Direction and size of the errors

The timetable errs in one direction: buses run ahead of it (median signed error +344 s at 10–20 minutes), and 62% of them pass more than two minutes before the published time. The official real time keeps that direction with less than half the size (+132 s): in about half of the cases at 10 to 30 minutes, the bus passes more than two minutes before the announced time. Bide+ is nearly centred (−24 s at 10–20 minutes) and is early by more than two minutes in 15–16% of cases at those horizons (Figure 3).

Table 4: Share within one minute (%) and median signed error (s) by time to the actual passage. A positive signed error means that the bus arrives earlier than predicted.

	Timetable		Official		Bide+	
Minutes	±1	Signed	±1	Signed	±1	Signed
0–3	10	+197	68	-10	70	-3
3–6	9	+230	28	-19	55	-7
6–10	9	+267	20	+35	47	-9
10–20	7	+344	14	+132	35	-24
20–30	7	+402	10	+188	25	-60
All	8	+313	22	+54	41	-18

Table 5: Inference. Columns 2–3: 95% bootstrap intervals (2,000 resamples) of the difference Bide+ minus official in the share within two minutes (percentage points), resampling trips or stops. Columns 4–6: mean loss differential $d = |e_{\text{off}}| - |e_{\text{B+}}|$ (s; positive favours Bide+), its standard error clustered by stop, and $z = d/\text{SE}$.

Minutes	By trip	By stop	d (s)	SE	z
0–3	$[-2.4, -0.8]$	$[-6.6, +3.0]$	-122	28	-4.4
3–6	$[+22.7, +25.4]$	$[+17.1, +30.3]$	-62	25	-2.5
6–10	$[+33.0, +36.4]$	$[+29.2, +39.1]$	-8	22	-0.4
10–20	$[+31.8, +35.9]$	$[+30.8, +36.7]$	93	17	5.4
20–30	$[+24.4, +29.7]$	$[+23.6, +30.5]$	160	25	6.5
All	$[+25.1, +28.2]$	$[+23.3, +29.5]$	49	15	3.3

Bide+ has, however, more very large errors at short horizons (Table 7): with the bus less than six minutes away, 2.4–2.6% of its predictions are more than ten minutes off, against 0.3–1.4% for the official real time. They occur both with a live position and in its fallback state (`estimated`), and about half of them at one Bilbao stop (Zabalburu, 305). These errors explain why its mean absolute error is larger than the official one below six minutes, even though it is within two minutes more often. From six minutes on, the official real time has both more moderate and more large errors.

6.6. Consequences for a passenger

Table 8 applies the passenger model to the pairs with the bus 3 to 30 minutes away, when a passenger decides when to leave for the stop. A passenger who arrives two minutes before the predicted time misses the bus in 61% of cases when following the timetable, 46% with the official real time and 14% with Bide+, and, when not missing it, waits on average 7.5, 4.0 and 3.3 minutes. Arriving earlier reduces the misses at the cost of longer waits, and the ranking of the three sources holds for every margin.

Table 6: Errors of the three sources by the waiting time announced by the official service (minutes). Columns as in Table 3.

		Timetable			Official real time			Bide+		
Announced	Pairs	Med.	±2	Early	Med.	±2	Early	Med.	±2	Early
0–2	19,362	401	17	45	52	78	0	44	82	5
3–5	15,120	390	19	47	92	61	5	56	77	6
6–9	16,694	386	20	51	116	51	21	60	76	8
10–19	39,750	422	17	57	178	35	41	81	66	10
20–29	30,750	482	16	68	261	25	63	106	56	16
30+	18,482	915	6	91	686	2	98	130	47	31

Table 7: Mean absolute error (s) and share of predictions more than ten minutes off (%), by time to the actual passage.

	Timetable		Official		Bide+	
Minutes	MAE	>10 min	MAE	>10 min	MAE	>10 min
0–3	733	37.1	60	0.3	182	2.4
3–6	743	37.9	136	1.4	198	2.6
6–10	751	39.4	208	5.3	216	2.8
10–20	771	42.8	314	14.2	222	3.0
20–30	808	46.0	413	23.6	252	3.9

6.7. Four buses drawn at random

Figure 4 shows the minute-by-minute predictions for four passages drawn at random. They illustrate the pattern of the aggregate results: the timetable is offset by a roughly constant amount, the official real time is closer but tends to stay too long, and Bide+ is the closest to the actual waiting time over most of the 30 minutes.

6.8. Stops, time of day and days

The ranking is not confined to a few stops, hours or days. Bide+ is more accurate than the official real time at 58 of the 59 stops (Figure 5), at every hour and on each of the five normal days (Figure 6, Table 10), at the priority stops and at the others, on weekdays and on Sunday, and for the second announced bus (Table 9). Fewer pairs are available in the evening because Thursday has no official times after 17:12 and evening services are less frequent.

6.9. Ablation: Bide+ without its learned segment times

To test whether the learned segment travel times explain the result, the same pairs were recomputed with Bide+ using empty tables: the vehicle is

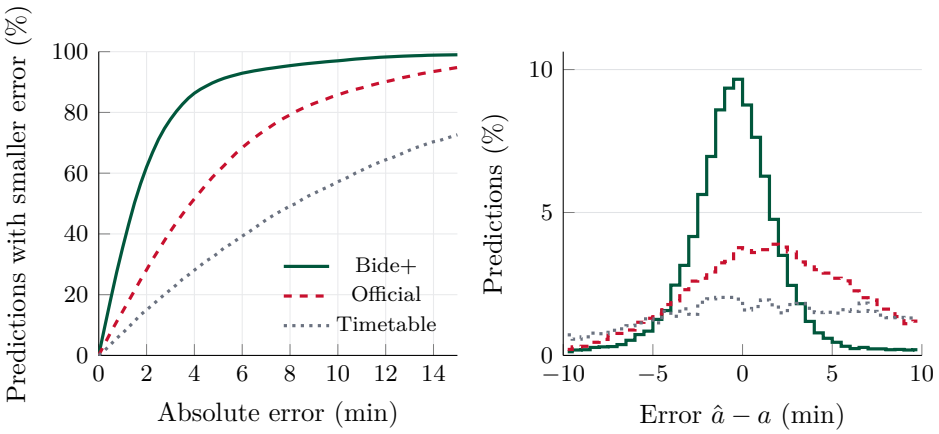


Figure 3: Errors with the bus 10 to 20 minutes away. Left: cumulative distribution of the absolute error. Right: distribution of the signed error in 30-second bins; positive means the bus arrives earlier than predicted. Beyond +10 minutes lie 37.6% of timetable, 13.4% of official and 2.2% of Bide+ predictions; below −10 minutes, 5.2%, 0.8% and 0.8%.

Table 8: Passenger model, bus 3 to 30 minutes away: a passenger reaches the stop k minutes before the predicted arrival. *Miss*: share of cases in which the bus has already passed (%). *Wait*: mean wait when it has not (min).

	Timetable		Official		Bide+	
k	Miss	Wait	Miss	Wait	Miss	Wait
1 min	65	7.2	54	3.5	24	2.7
2 min	61	7.5	46	4.0	14	3.3
3 min	58	7.9	39	4.4	9	4.1

still located in timetable time, but the arrival is built from the timetable differences only (Table 11). Without learning, Bide+ is within two minutes in 32% of cases, below the official real time (39%); with learning, in 66%. The learned segment times are therefore what makes Bide+ more accurate than the operator, and locating the bus alone is not enough.

6.10. Sensitivity

Table 12 repeats the comparison under 17 alternative choices. The ranking of the three sources is unchanged in all of them. Shifting the actual passage later by 60 s, the least favourable case for Bide+, lowers its share within two minutes from 66% to 57% and the official one from 39% to 37%. A stricter identity check raises both (47% and 70%). Including the pairs in which Bide+ could only give its fallback prediction, and dropping the

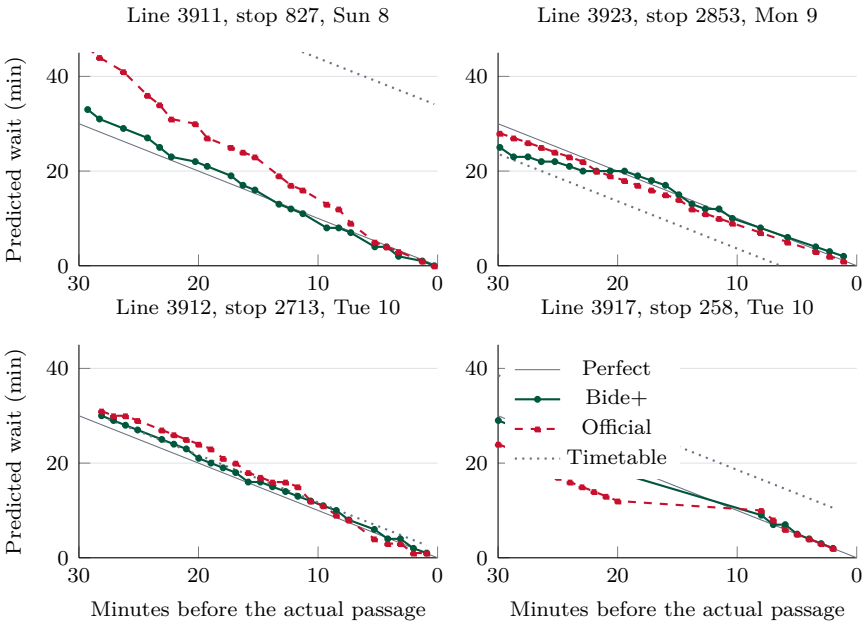


Figure 4: Predicted waiting time against the actual one for four buses drawn at random (fixed seed) among the passages with queries from at least 25 minutes before. The thin line is a perfect prediction.

identity check, gives 36% and 63%. Keeping one random pair per passage, which removes the dependence between consecutive minutes, gives 43% and 68%.

6.11. The strike day

On Wednesday 11, with minimum services, the ranking was the same as on normal days (Table 10). Bide+ was more often early than on normal days: its learned travel times come from normal traffic, and the buses ran faster than learned.

7. Discussion

7.1. What the comparison shows

On the buses compared, the published timetable is a poor guide to when the bus will come, the operator's real time removes part of its error, and Bide+, built only on the operator's open data, is within two minutes of the actual passage more often than the operator's real time from three minutes

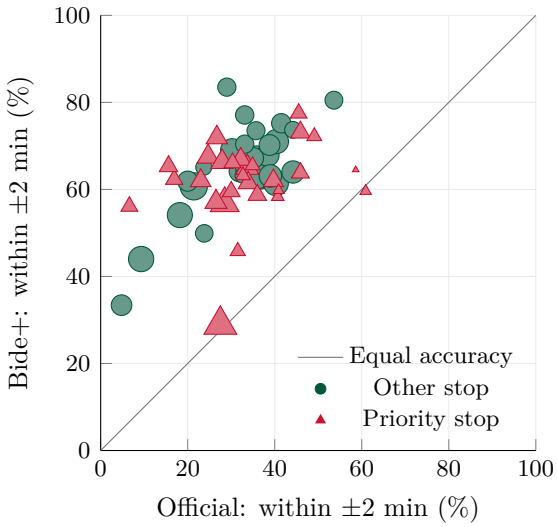


Figure 5: Each point is one stop (bus 3 to 30 minutes away; marker area proportional to the number of pairs). Bide+ is more accurate at 58 of the 59 stops, and at 56 the 95% interval of the difference (by trip) excludes zero.

before arrival onwards. The gap is largest, about 34 percentage points, with the bus 6 to 20 minutes away, the range in which a passenger decides when to leave for the stop. Below three minutes the official real time is slightly more accurate, and Bide+ has a small share of very large errors at short horizons that the operator does not have. Those errors are concentrated at one stop, and they point to where the method can be improved.

7.2. Possible explanations

The ablation (Table 11) identifies the learned segment travel times as the component that makes Bide+ more accurate: without them it falls below the official real time. This is consistent with the timetable on this network running behind the buses, and with the official real time keeping part of that bias (Figure 3); a predictor that learns how long each segment actually takes corrects it. How the official service computes its waiting times is not published, so this paper cannot say why its errors take the form they do. Testing other components of Bide+ (pace factors, per-trip corrections, the fallback layers) is left for future work.

7.3. Implications

Operators that publish vehicle positions make possible independent predictions that, on this network and week, were more accurate than their own.

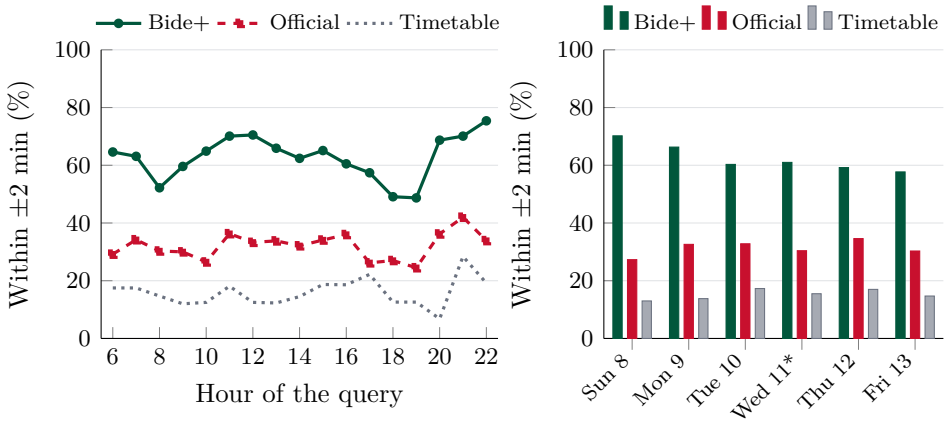


Figure 6: Share within two minutes with the bus 3 to 30 minutes away. Left: by hour of the query (five normal days; fewer pairs after 17:00, Table 1). Right: by day (* general strike, not part of the main results).

Table 9: Errors by group, all horizons up to 30 minutes. Columns as in Table 3.

		Timetable			Official real time			Bide+		
Group	Pairs	Med.	±2	Early	Med.	±2	Early	Med.	±2	Early
Weekdays	117,221	442	16	57	165	40	39	80	64	13
Sunday	22,937	774	13	75	212	34	55	70	73	16
Priority stops	54,602	520	16	62	176	39	41	84	63	14
Other stops	85,556	455	16	59	166	39	42	74	68	13
Next bus (e_1)	134,086	477	16	60	168	39	41	77	66	13
Second bus (e_2)	6,072	494	13	71	210	30	52	106	55	14

The most direct implication is for evaluation: operators that also publish their predictions, for example as GTFS-Realtime trip updates or SIRI estimated timetables, would allow their accuracy to be audited independently and continuously, as done here by recording a web service. Journey planners that show the operator's figures unchanged inherit their accuracy; we have not measured any of them.

7.4. Limitations

- **Coverage.** Only 16% of the official announcements of the next bus less than 30 minutes ahead entered the main analysis (22.7% could be paired at all). The results describe buses tracked in the feed, mostly in the middle of their trip, whose identity could be confirmed, not every announcement.

Table 10: Errors by day, all horizons up to 30 minutes. Columns as in Table 3. The strike day is shown for reference and is not part of the main results.

		Timetable			Official real time			Bide+		
Day	Pairs	Med.	±2	Early	Med.	±2	Early	Med.	±2	Early
Sun 8 Nov	22,937	774	13	75	212	34	55	70	73	16
Mon 9 Nov	27,866	458	14	59	163	40	40	71	69	13
Tue 10 Nov	33,655	458	18	58	165	40	39	82	64	11
Wed 11 Nov (strike)	21,372	562	16	66	180	38	48	82	65	26
Thu 12 Nov	29,654	414	17	56	156	42	37	82	63	11
Fri 13 Nov	26,046	448	15	56	178	37	41	88	61	15

Table 11: Ablation: share within two minutes (%) on the same pairs, with Bide+ without and with its learned tables.

Minutes	Pairs	Timetable	Official	Bide+ without learning	Bide+
0–3	17,125	19	92	81	90
3–6	17,530	18	55	46	79
6–10	21,869	17	39	32	74
10–20	47,469	15	28	22	62
20–30	36,165	14	21	15	48
All	140,158	16	39	32	66

- **Ground truth from the same feed.** The actual passage is reconstructed from the feed field that Bide+ also uses. It was not validated against an independent source such as field observation or door records. Shifting it by up to 60 s does not change the ranking, but a validation in the field remains necessary.
- **Matching.** The official service does not identify the vehicle it announces. The distance check removes most mismatches, and stricter or no checks do not change the ranking, but some may remain.
- **Selection of stops.** The stops were chosen by the author; 28 of them partly because the timetable deviates most there. Results without them are similar, but the stops are not a random sample of the network.
- **One week, one network.** Five normal days in November, one of them a Sunday. Day-level variance cannot be estimated, and other seasons, networks or operators may differ.
- **Replay, not the app.** Bide+ was recomputed with its reference implementation and frozen tables; the app, which recalibrates every

Table 12: Sensitivity analyses: share within two minutes (%) over all horizons and with the bus 10 to 20 minutes away.

		All horizons			10–20 min	
Variant	Pairs	Timet.	Official	Bide+	Official	Bide+
Main analysis	140,158	16	39	66	28	62
Actual passage −60 s	140,158	16	38	64	26	61
Actual passage −30 s	140,158	16	39	66	27	63
Actual passage +30 s	140,158	15	39	63	29	59
Actual passage +60 s	140,158	15	37	57	29	52
Official +30 s (minutes truncated)	140,158	16	39	66	27	62
Bide+ in seconds	140,158	16	39	66	28	63
Bide+ minutes truncated	140,158	16	39	63	28	58
Stricter identity check	81,393	17	47	70	32	63
No identity check	158,567	16	37	65	27	62
No identity check, with fallback	200,383	18	36	63	29	61
With stop 3394 and impossible passages	141,578	16	39	65	28	62
One pair per passage (random)	7,925	18	43	68	27	62
Without priority stops	85,556	16	39	68	28	63
Bide+ with live position only	127,158	15	39	68	29	64
Threshold ±1 min	140,158	8	22	41	–	–
Threshold ±3 min	140,158	23	52	79	–	–
Mean absolute error (s)	140,158	769	270	221	–	–

night, was not measured.

- **Analytical choices.** The scoring rule, horizons and matching follow the companion paper and were fixed before the analysis, but the study was not preregistered. A confirmatory replication on another week, ideally by a third party, would strengthen the conclusions.

8. Conclusions

For one week, at 59 stops of the Bizkaibus network, the official real-time waiting times were recorded minute by minute and compared, bus by bus, with the published timetable and with the arrivals predicted by the Bide+ method from the operator's open position feed and timetable. Over all horizons up to 30 minutes, the timetable was within two minutes of the actual passage in 16% of the paired cases, the official real time in 39% and Bide+ in 66%. The official real time was slightly more accurate with the bus less than three minutes away; from three minutes on Bide+ was more accurate, at almost every stop, hour and day, and an ablation attributes the difference to its learned segment travel times. These results apply to the buses whose passage could be observed and whose identity could be

confirmed, and they call for an independent validation of the ground truth and a replication on other periods and networks.

Declaration of competing interest

The author designed the Bide+ method and develops the independent Bide+ passenger information app that uses it, and therefore has an interest in its results. To limit the effect of this interest on the evaluation, the method was frozen before the test week, its predictions were recomputed with the public reference implementation, the analysis rules are stated in Section 5, and all data and code are published so that the comparison can be repeated independently. The author has no relationship with Bizkaibus, the Diputación Foral de Bizkaia or the operators of the service.

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CRediT authorship contribution statement

Yeray Garrido: Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing.

Data availability

The capture program, the list of stops with the reason for each, the days and stops used and excluded, the comparison, the tables and the figure data are in the folder `articulo-comparacion` of the repository `https://github.com/Garridoparrayeray/bide-eta` (code under the MIT licence, texts and derived data under CC BY 4.0). The raw captures (feed copies, official responses and the timetable of each day) are assets of the release `comparacion-2026-11` of the same repository. An archived copy with a DOI will be deposited on Zenodo on acceptance. The position feed and the timetable are open data published by the Diputación Foral de Bizkaia and listed in the Spanish National Access Point (Ministerio de Transportes y Movilidad Sostenible 2026).

Data collection and terms of use

The official waiting times were obtained from the public web service used by the official Bizkaibus app, which requires no authentication. The capture queried it sequentially, about one request per second in total, with a user agent that identifies the project, and only during service hours. No permission was sought from the operator. The data contain no personal information: only stop, line and vehicle identifiers. The raw responses are redistributed for research with attribution, and they will be removed if the publisher of the data asks.

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